# Definitional-Closure-Constrained Multitask Retrieval of Leaf LMA, EWT, and FMC From Hyperspectral Spectra

Jinlong Liu, Xuanwen Wang, and Quan Wang

***Abstract*—Leaf mass per area (LMA) and equivalent water thickness (EWT) are characteristics of leaf structure and water status. The ratio of these two traits, fuel moisture content (FMC), reflects the balance between water and structural components, involving a series of ecological processes. Simultaneously retrieving these traits from hyperspectral reflectance should facilitate the disentangling of water and structural components, enabling per-sample physical consistency checks via definitional closure. This enhances the credibility of cross-species and cross-environment interpretations compared to traditional single-trait or stepwise retrievals. In this study, we propose a physically consistent, multitask framework based on a temporal convolutional network (TCN) that uses a shared spectral encoder to learn synergistic information across three traits. The framework uses a soft-physics formulation of "physical baseline + residual correction" to unify predictive accuracy and physical self-consistency. Experiments on the composited EcoSIS dataset demonstrate stable retrieval performance for all three traits (test-set $R^2 \approx 0.89$–$0.93$) and show that three-task joint learning significantly outperforms stepwise-based derivation (derived FMC root mean square error (RMSE) $\approx$ 47.63%). Physics regularization reduces the symmetric relative difference (SRD)—a measure of internal physical inconsistency—from ~4.71% to ~0.45%, effectively mitigating the "high-accuracy illusion" of unconstrained models. The model maintains stable accuracy and consistency across spectral resolutions ranging from 1 to 50 nm. These results suggest that the proposed framework could serve as a general retrieval paradigm for composite traits with explicit analytics and provide a more interpretable and transferable solution for generalized leaf-trait retrieval across sensors and ecological domains.**

***Index Terms*—Hyperspectral, joint retrieval, multitask learning (MTL), physical consistency, soft physical constraints, temporal convolutional network (TCN).**

## I. Introduction

LEAF functional traits provide an entry point for observing and measuring the structure and processes of ecosystems. They also play a key role in connecting spectral information to ecological processes and allowing us to understand them in a mechanistic way [1]. Among the many leaf functional attributes, leaf mass per area (LMA) and equivalent water thickness (EWT) are key traits that describe leaf structural characteristics and water balance, respectively, and influence leaf optical performance as well [2], [3], [4], [5]. Together, LMA and EWT regulate critical spectral responses, including leaf light scattering and water absorption features in the shortwave infrared. These traits are closely associated with vegetation growth conditions and stress-adaptation capacity [6], [7]. Furthermore, the ratio of EWT to LMA, known as fuel moisture content (FMC), serves as an integrated quantitative indicator of vegetation water status. It has been widely used in wildfire risk assessment and prediction [8]. FMC also captures drought-induced water deficits, changes in leaf physiological activity, and potential vegetation vulnerability, which are closely linked to water–carbon coupling processes in terrestrial ecosystems [9]. Importantly, FMC is not an independent physiological variable; rather, it is a composite ratio trait that has an algebraically linked relationship with EWT and LMA through its definitional closure.

Hyperspectral remote sensing offers rich and continuous spectral information for retrieving leaf traits [10], [11], [12], particularly in the shortwave infrared region, where clear absorption features related to water and dry matter can be observed [13], [14]. Traditional mechanistic inversion usually uses radiative transfer models (RTMs) to estimate parameters. While these methods are clearly physically interpretable, they are often limited by illposedness and parameter identifiability, which should lead to significant uncertainty in the results [15], [16]. Meanwhile, conventional statistical learning methods have long served as empirical baselines for hyperspectral trait retrieval, with advantages such as simplicity of implementation and relatively strong interpretability. However, they are often subject to limitations under conditions of strong nonlinear coupling, cross-dataset transfer, and complex noise [17]. A notable shift toward deep learning has been observed in recent studies. For example, large-scale deep-learning studies have reported improved robustness and accuracy in retrieving key leaf traits, such as EWT and LMA, under complex hyperspectral conditions [18], while multitrait convolutional neural network (CNN) frameworks have outperformed single-trait CNNs and advanced PLSR models across different vegetation types and sensors, suggesting that simple empirical methods may be

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Jinlong Liu is with the Graduate School of Science and Technology, Shizuoka University, Shizuoka 422-8529, Japan.

Xuanwen Wang is with the United Graduate School of Agriculture Science, Gifu University, Gifu 501-1193, Japan.

Quan Wang is with the Faculty of Agriculture, Shizuoka University, Shizuoka 422-8529, Japan (e-mail: wang.quan@shizuoka.ac.jp).

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insufficient when trait relationships are strong and spectral conditions are highly variable [19]. These results prove that deep neural networks are a promising approach for leaf-trait retrieval. However, purely data-driven deep learning models largely function as "black boxes" and may overlook physical trait relationships. Consequently, even when these models demonstrate good statistical performance, they may produce predictions that are physically or physiologically implausible. Therefore, the central issue addressed in this study is not whether deep learning can surpass simple empirical baselines, as demonstrated in earlier research, but rather, how we can incorporate definitional closure into a multitask deep-learning framework to enhance the precision and physical consistency of composite-trait retrieval.

The lack of physical interpretability in data-driven models has led to increased attention being paid to physics-informed or physics-aware machine learning. This learning paradigm explicitly embeds physical knowledge into machine-learning models [20], [21]. In vegetation optical remote sensing, these ideas are often carried out using a combination of RTMs and statistical or deep-learning techniques, or by incorporating differentiable RTMs as forward modules within neural networks to allow for semisupervised inversion and ensure physical consistency [22], [23]. Physics-informed neural networks (PINNs) exemplify a prominent class of such methodologies, wherein residuals of the governing equations, in combination with boundary and initial conditions, are integrated into the loss function [20]. PINNs are particularly well-suited to problems in which partial differential equations constitute the primary prior knowledge. However, previous studies have shown that mismatched convergence rates among different loss components in PINN training can induce optimization pathologies, leading to training instability or a failure to converge [21]. More importantly, the connection between spectra and vegetation characteristics is encoded in RTMs, and incorporating RTMs within network designs still faces the basic problem that different parameter combinations can produce similar spectral signatures. As a result, inversion solutions are often nonunique and highly sensitive to noise, resulting in severe parameter identifiability issues [24].

For the relationships among LMA, EWT, and FMC considered in this study, their definitional algebraic closure provides the most robust and directly testable source of physical information. When only a single trait is retrieved [19], the coupled spectral effects of water and structural components may cause attribution ambiguity: observed spectral differences may not necessarily correspond to a single physiological change, thereby increasing uncertainty in interpreting FMC variations. Although a stepwise strategy–retrieving LMA and EWT first and then computing FMC—formally satisfies the defining equation [25], it remains unclear whether errors will accumulate and be amplified through the ratio operation under observation noise, model bias, or mismatches in sample distributions, thereby compromising the stability and credibility of FMC estimates. Therefore, it is important to investigate the feasibility of jointly modeling LMA, EWT, and FMC at a consistent observation scale. This study investigates whether such joint modeling can maintain predictive accuracy, improve result credibility, and enhance cross-species and cross-environment interpretability. Such analytical relationships are incorporated as explicit constraints or soft regularization terms that penalize inconsistency, or are injected into models via structured designs such as a "physical baseline + residual correction" framework. This strategy avoids the identifiability and computational burden caused by full RTM hard constraints, while physical consistency can be directly measured using closure errors and further used for model diagnosis [26].

Following this background, this study uses multitask learning (MTL) to jointly retrieve LMA, EWT, and FMC within a unified framework. In this framework, a shared spectral-domain encoder learns common representations, and task-specific regression heads capture task-dependent differences. This structure helps to improve model generalization and reduce the risk of overfitting. To address challenges in multitask training, such as gradient conflict/alignment and the difficulty calibrating weights, this study explores the applicability of equal-weighting [27] and adaptive weighting strategies (using Nash-MTL as an example) [28], [29], and systematically evaluates both predictive accuracy and physical consistency. For hyperspectral modeling, compared with fully connected architectures that treat spectra as unordered features, 1-D convolution better conforms to spectral continuity through local receptive fields and parameter sharing [30], and can help suppress noise and improve sample efficiency [18]. On this basis, we employ a temporal convolutional network (TCN)-inspired 1-D convolutional backbone [31]. While TCNs were initially developed for sequence modeling, their ability to handle hyperspectral reflectance data along wavelengths makes them well-suited for spectral modeling as well. In our case, we use the TCN-style design to model dependencies along the spectral dimension rather than temporal dynamics. This approach is supported by related hyperspectral studies showing that temporal-convolution-style or dilated residual convolutional designs can outperform standard 1-D or conventional convolutional baselines by capturing a wider spectral context more effectively [18], [32]. In particular, dilated convolutions expand the receptive field and allow the model to capture long-range dependencies among spectral bands efficiently, without greatly increasing model complexity, while residual connections facilitate deeper feature extraction and improve training stability [33]. In addition, the fully convolutional structure enables parallel computation, making it well-suited for end-to-end inversion of high-dimensional hyperspectral sequences. The main contributions of this study are as follows.

1) We develop a TCN-based multitask joint inversion model to simultaneously estimate LMA, EWT, and FMC, and systematically compare single-task, dual-task, and three-task configurations, as well as equal weight and Nash adaptive weighting strategies, to assess their differences in accuracy, stability, and generalization performance.
2) We propose a soft physics-consistency constraint tailored to the trait definitional relationship, embedding the closure relation into the training objective in a

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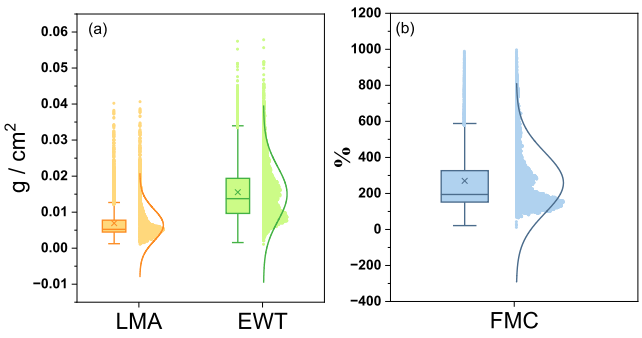



Fig. 1. Dataset description. (a) LMA, EWT. (b) FMC.

regularization form. This enables quantification of the model's rate of physical violation while keeping model complexity under control.

3) We evaluate retrieval accuracy and physical consistency under different spectral resolution settings and analyze the role and potential transferability of the soft physical constraint in cross-scale applications, thereby providing a basis for developing generalizable and interpretable vegetation trait retrieval methods.

## II. Materials and Methods

### A. Experimental Dataset

In this study, we selected nine mutually independent subsets from the [ecological spectral information system (EcoSIS), https://ecosis.org/] spectral library [34], [35], [36], [37], [38], [39], [40], [41], [42] and aggregated them into a unified database to support the joint retrieval of leaf traits. The samples are mainly distributed across North America and Europe. A smaller number of samples are from East Asia and Australia. This distribution covers diverse terrestrial ecosystems and plant functional types (PFTs), enabling the evaluation of model generalization across ecological contexts. The final processed data and processing scripts are available in a public repository (https://doi.org/10.6084/m9.figshare.31245604). The aggregated database contains 7393 records. Each record includes a 400–2400-nm reflectance spectrum (original sampling interval: 1 nm) and corresponding leaf trait measurements (LMA, EWT, and FMC). Vegetation types include evergreen needleleaf forest, evergreen broadleaf forest, deciduous needleleaf forest, deciduous broadleaf forest, shrubland, grassland, and lianas, among others. To reduce the influence of differences in measurement and reporting protocols among the subsets, we applied a preprocessing procedure (see Section II-B). The distributions of leaf traits are shown in Fig. 1, with LMA ranging from 0.012 to 0.0408 g/cm$^2$, EWT ranging from 0.0015 to 0.0580 g/cm$^2$, and FMC ranging from 21.2042% to 999.4299%.

### B. Data Preprocessing

To minimize random noise in hyperspectral data while maintaining the shape of the absorption features as accurately as possible, we applied Savitzky–Golay smoothing along the wavelength dimension [43]. Within a sliding window, we fit a polynomial by least squares and take the fit value at the window center as the smoothed output. We set the parameters to a window length of 15 and a polynomial order of 3. Spectral resampling was performed using a mean-binning strategy [44]. Starting from the minimum wavelength, we partitioned the spectrum into consecutive intervals according to the target resolution. Then, we used the mean of the original bands within each interval as the representative value for that bin. To mitigate the additive and multiplicative effects introduced during sample measurements—such as variations in illumination intensity, probe contact, and scattering differences—we applied a samplewise standard normal variate (SNV) transformation to each spectrum [45]. For a single-sample spectral vector $\mathbf{x}$

$$\mathbf{x}^{(snv)} = \frac{\mathbf{x} - \mu(\mathbf{x})}{\sigma(\mathbf{x})} \tag{1}$$

where $\mu(\mathbf{x})$ and $\sigma(\mathbf{x})$ denote the mean and standard deviation of the spectrum for that sample across all wavelengths, respectively.

### C. Model Structure

In this study, we developed a multitask regression model based on a 1-D dilated TCN for the hyperspectral retrieval of leaf traits. We incorporated Nash-based adaptive Nash-MTL to dynamically balance gradient conflicts among tasks. We also introduced a physics-based consistency constraint related to leaf water status, deriving FMC from LMA and EWT, to enhance model interpretability and generalization performance. To address the skewed distributions of the target variables and ensure positive predictions, the targets were log-transformed before model training. All evaluation metrics were computed on the original scale.

*1) Multitask Sequential Convolutional Network Structure:* Building on the TCN architecture, this study developed a multitask 1-D convolutional network for hyperspectral signals (Multitask TCN). The network consisted of a shared feature-extraction backbone and task-specific regression heads, following the hard parameter sharing paradigm commonly used in MTL to exploit task relatedness and improve generalization (Fig. 2) [46]. The shared backbone began with a multiscale convolutional stem: the input was fed into three parallel 1-D convolution branches (kernel sizes 9, 19, and 39), whose outputs were concatenated along the channel dimension and then fused by an additional convolution to form a joint representation of narrowband absorption features and broadband spectral shape [47]. The multiscale convolutional stem was mainly used to capture local spectral features at different scales. Based on these features, the TCN module further combined multiscale information and learned broader dependencies along the spectral dimension, helping capture long-range relationships between bands that are difficult to model with standard convolutions alone. The main backbone was constructed by stacking multiple temporal convolutional residual blocks. Each residual block contained two 1-D dilated convolutions, combined with batch normalization and rectified linear unit (ReLU) activation to stabilize training [48]. The dilation rates increased progressively as 1, 2, and 4, expanding the receptive field under a controlled parameter budget and capturing long-range interband dependencies

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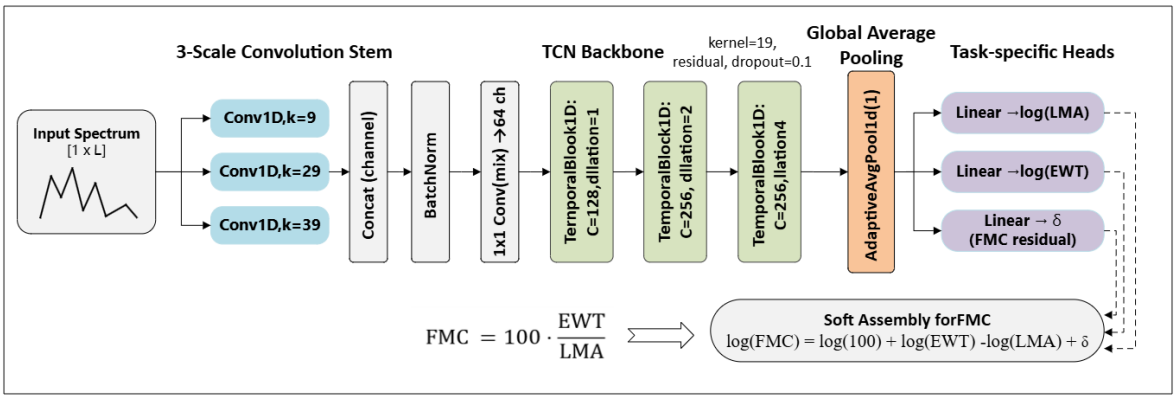



Fig. 2. Illustration of the model structure.

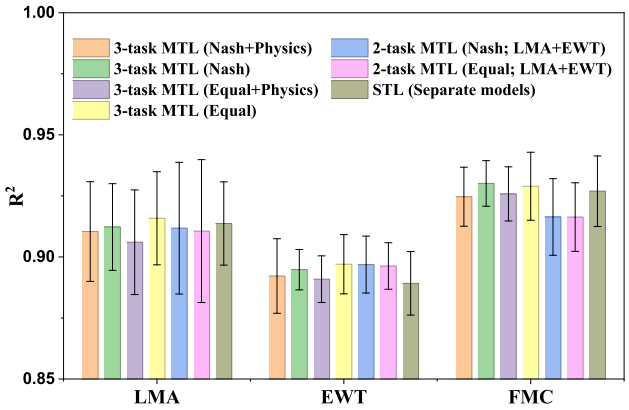



Fig. 3. $R^2$ across different ablation settings. Error bars represent the standard deviation of three independent experiments.

[49]. Residual connections were used to alleviate degradation in deep networks; when channel dimensions mismatch, a $1 \times 1$ convolution was applied to the shortcut path for alignment [50]. During training, dropout was employed to reduce overfitting [51]. Finally, the backbone outputs were compressed into a sample-level feature vector via global average pooling [52], and an independent linear regression head was assigned to each task to produce the final predictions.

*2) Physical Consistency Constraints and "Soft-Physics" FMC Modeling:* For each task $t$ (LMA, EWT, and FMC), let $y_t$ denote the raw target and $z_t = \log(y_t)$ the transformed target used for optimization. The regression loss $\mathcal{L}_t$ was defined using the Huber loss

$$\mathcal{L}_t = \text{Huber}\left(\hat{z}_t, z_t; k\right) \tag{2}$$

where $\hat{z}_t$ and $z_t$ are the predicted and observed values in the log-transformed space, respectively, and $k$ is the Huber transition parameter. The Huber loss was adopted because it provides a robust compromise between sensitivity to small residuals and resistance to moderate outliers in the aggregated EcoSIS dataset [53], [54].

At the blade scale, the three satisfied the following analytical closure relationship. In the multitask setting, the network was no longer trained to directly regress FMC. Instead, we adopted a soft-physics formulation of "physical baseline + residual correction" and predict a correction term $\delta$. In particular, the model first obtained LMA and EWT from the first two heads (in log space, $\log\widehat{\text{EWT}}$ and $\log\widehat{\text{LMA}}$), and then computed the baseline prediction $\widehat{\text{FMC}}_{\text{base}}$ as

$$\text{FMC} = 100 \cdot \frac{\text{EWT}}{\text{LMA}}$$
$$\log\left(FMC_{\text{base}}\right) = \log\left(100\right) + \log EWT - \log LMA. \tag{3}$$

The third task header did not directly output a $\log(\widehat{\text{FMC}}_{\text{base}})$, but instead output the residual correction term $\delta$ for additive correction

$$\log(\widehat{\text{FMC}}) = \log(\widehat{\text{FMC}}_{\text{base}}) + \delta. \tag{4}$$

In the loss function, we further imposed a "shrink-to-zero" regularization on $\delta$, encouraging the model to follow the analytical closure baseline when evidence is insufficient, while still allowing adaptive residual corrections in the presence of systematic biases (e.g., measurement noise, definitional inconsistency, or scale effects). This was implemented as a Huber loss between $\delta$ and 0

$$\mathcal{L}_{\text{cons}} = \text{Huber}(\delta, 0). \tag{5}$$

The final total loss was

$$\mathcal{L} = \sum_t \lambda_t \mathcal{L}_t + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} \tag{6}$$

where $\lambda_t$ denotes the weight of task $t$ (which can be combined with adaptive weighting schemes), and $\lambda_{\text{cons}}$ is the consistency-regularization coefficient implemented via a staged warm-up schedule: it was set to zero for the first 50 epochs and then linearly increased to its maximum over the next 50 epochs. To balance the goals of high retrieval accuracy and strict physical closure, $\lambda_{\text{cons}}$ was set to 0.03.

*3) Multitask Weighting and Task Loss Functions:* In the MTL setting, this study explored Nash-MTL [29], in addition to equal weighting [27], to adaptively update the task weights $\alpha = [\alpha 1, \alpha 2, \alpha 3]$. In particular, within each minibatch, we computed the gradient vectors of each task loss with respect to the shared parameters. Then, we flattened them to construct a gradient similarity matrix. Finally, we iteratively solved for $\alpha$ via projected gradient descent, such that the aggregated gradient reached a fairer or more balanced Nash equilibrium across tasks [29].

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### D. Training Strategy and Evaluation

Data partitioning followed a hybrid scheme combining a fixed hold-out test set with $K$-fold cross-validation [55]. Specifically, 15% of the samples were first split off as a hold-out test set shared across all folds, while the remaining data were used for fivefold cross-validation for training and validation. Model training employed the AdamW optimizer (lr = $1 \times 10^{-3}$, weight_decay = $1 \times 10^{-4}$) with a batch size of 64 and a maximum of 200 epochs. For validation monitoring, an "average-performance-first" criterion was adopted: the mean multitask validation coefficient of determination ($R^2$) across tasks served as the monitoring metric for early stopping and learning-rate scheduling. For final accuracy evaluation, predictions were transformed back to the original (nonlog transformed) scale, and two metrics were used: root mean square error (RMSE) and the $R^2$, defined as

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{7}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{8}$$

where $y_i$ and $\hat{y}_i$ denote the observed and predicted values of a target trait, respectively, $\bar{y}$ is the mean of the observed values, and N is the number of samples.

To quantify deviations of FMC from the analytical baseline, the symmetric relative difference (SRD) [56] was additionally computed as a physical-consistency diagnostic metric

$$SRD = \frac{1}{N}\sum_{i=1}^{N}\frac{2|\widehat{FMC}_i - FMC_{phy,i}|}{|\widehat{FMC}_i| + |FMC_{phy,i}|} \times 100 \tag{9}$$

where $\widehat{FMC}$ denotes FMC directly predicted by the model, and $FMC_{phy}$ is computed from the predicted EWT and LMA via the definitional equation. A lower SRD indicates better agreement between the model-predicted FMC and the physically derived baseline, and therefore reflects stronger physical consistency (i.e., a lower degree of physical violation).

All subsequent analyses were based on results obtained from three independent runs of each model with different random seeds. In each run, the dataset split was kept fixed, ensuring that all compared models used identical training, validation, and test sets. The models were implemented using the PyTorch framework. Experiments were conducted on a server equipped with an Intel[1] Core[2]i9-14900KF CPU (4.96 GHz), 128 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB).

## III. Results

### A. Accuracy Assessment Across Different Ablation Settings

This study used a comprehensive, multispecies, spectral-trait dataset to systematically evaluate the influence of different model configurations (single-task learning versus MTL), weighting strategies (Nash adaptive versus equal), and physics-regularization constraints on retrieval accuracy. The mean test-set performance metrics of each scheme are summarized in Table I.

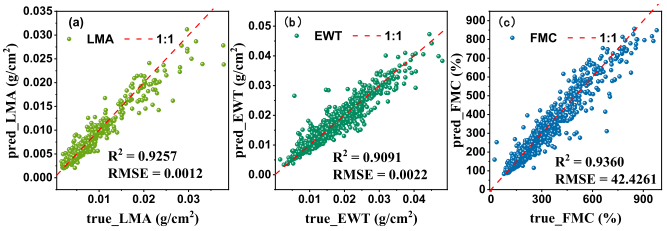



Fig. 4. Scatter plots of predicted versus observed values for (a) LMA, (b) EWT, and (c) FMC using the best-performing MTL strategy. The red dashed line indicates the 1:1 relationship.

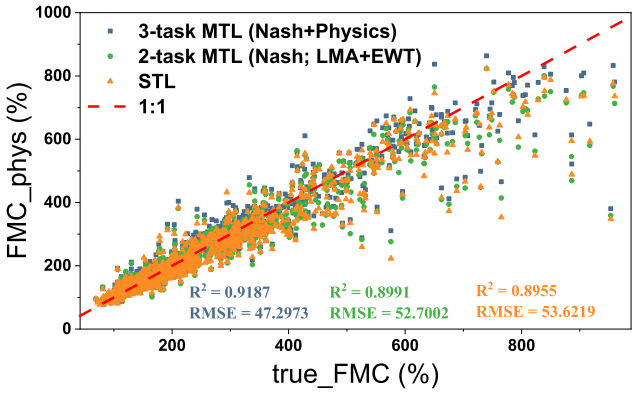



Fig. 5. Scatter plots of physically derived FMC (FMC_phys) versus observed FMC (True_FMC).

All models achieved high and stable accuracy for the retrieval of the fundamental traits LMA and EWT ($R^2$ ranges: 0.906–0.916 for LMA and 0.889–0.897 for EWT). The MTL strategy with equal weights (MTL with equal weights) produced the best overall performance with RMSE values of 0.0013 and 0.0024 g/cm$^2$ for LMA and EWT, respectively. On the contrary, Nash adaptive weighting did not yield a clear accuracy gain in these experiments (Table I; Fig. 3). This suggests that dynamically rebalancing gradients via adaptive weights was not substantially better than the simple equal-weight scheme under the current task setting. Furthermore, the equal-weight strategy demonstrated greater robustness across repeated runs.

For FMC retrieval, the three-task MTL framework markedly outperformed the two-task model that included only LMA and EWT. While the two-task model could accurately estimate the component traits, the FMC computed via the physical equation exhibited significant error accumulation, with the RMSE increasing to approximately 47.6%. By contrast, explicitly incorporating FMC as a learning target in the three-task setting effectively constrained the shared multitask representation, improving the error structure and substantially reducing FMC uncertainty. Interestingly, adding physics regularization to the three-task model slightly decreased the statistical accuracy of FMC ($R^2$ decreased from 0.9301 to 0.9246). This reflects the tradeoff between data-fitting capacity and physical consistency. The regularization term forces the solution space to converge toward physically interpretable regions. This

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TABLE I
QUANTITATIVE COMPARISON OF RETRIEVAL ACCURACY FOR LMA, EWT, AND FMC UNDER DIFFERENT MODELING STRATEGIES

| | LMA ($R^2$) | RMSE ($g/cm^2$) | EWT ($R^2$) | RMSE ($g/cm^2$) | FMC ($R^2$) | RMSE (%) |
|---|---|---|---|---|---|---|
| 3-task MTL (Nash) + Physics regularization | 0.9104 | 0.0014 | 0.8922 | 0.0025 | 0.9246 | 45.1872 |
| 3-task MTL (Nash) | 0.9123 | 0.0014 | 0.8948 | 0.0024 | **0.9301** | **43.5541** |
| 3-task MTL (Equal) + Physics regularization | 0.9060 | 0.0014 | 0.8909 | 0.0025 | 0.9258 | 44.8521 |
| 3-task MTL (Equal) | **0.9158** | **0.0013** | **0.8970** | **0.0024** | 0.9289 | 43.8374 |
| 2-task MTL (Nash weighting; LMA+EWT) | 0.9118 | 0.0014 | 0.8969 | 0.0024 | 0.9164 (computed) | 47.6002 |
| 2-task MTL (Equal weights; LMA+EWT) | 0.9106 | 0.0014 | 0.8963 | 0.0024 | 0.9163 (computed) | 47.6362 |
| STL (separate models) | 0.9137 | 0.0013 | 0.8892 | 0.0025 | 0.9269 | 44.4705 |

Note: because the 2-task MTL setting does not predict FMC directly, FMC under this strategy was computed from the predicted EWT and LMA to facilitate comparison. For each trait, $R^2$ and RMSE are reported as evaluation metrics.

TABLE II
QUANTITATIVE COMPARISON OF THE PHYSICS-DERIVED FMC INVERSION ACCURACY AND PHYSICAL CONSISTENCY UNDER DIFFERENT MODELING STRATEGIES

| | FMC_phys ($R^2$) | RMSE (%) | SRD (%) |
|---|---|---|---|
| 3-task MTL (Nash) + Physics regularization | 0.9221 | 45.9225 | 0.8552 |
| 3-task MTL (Nash) | **0.9286** | **44.0453** | 0.8351 |
| 3-task MTL (Equal) + Physics regularization | 0.9254 | 44.9812 | **0.4538** |
| 3-task MTL (Equal) | 0.9274 | 44.3198 | 0.8069 |
| 2-task MTL (Nash weighting; LMA+EWT) | 0.9164 | 47.6002 | — |
| 2-task MTL (Equal weights; LMA+EWT) | 0.9163 | 47.6362 | — |
| STL (separate models) | 0.9313 | 43.1278 | 4.7107 |

Note: FMC_phys refers to the FMC computed from predicted LMA and EWT equations. SRD (%) represents the Symmetric Relative Difference between directly predicted FMC and FMC_phys, indicating internal model consistency.

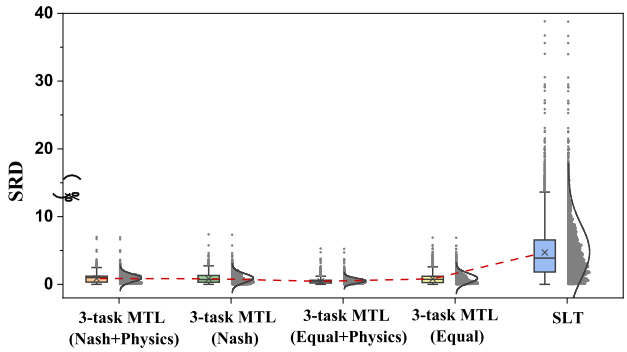



Fig. 6. Distribution of SRD across test samples. A narrower distribution closer to zero indicates higher physical self-consistency.

reduces the degree of freedom for overfitting the training data, but it can yield solutions with stronger physical meaning.

Fig. 4 shows representative scatter plots of the predicted versus observed values. For LMA and EWT, the data points are tightly clustered around the 1:1 line. However, mild heteroscedasticity is observed in the high-value range, indicating that extreme samples are a significant source of error. The FMC scatter distribution also shows good agreement, confirming again that joint inversion can improve estimation of the composite trait (FMC) while maintaining high accuracy for the component traits (LMA and EWT) under the MTL framework.

### *B. Evaluation of Physical Consistency and Model Self-Constraint*

Building on the accuracy assessment in Section III-A, we further investigate—along two complementary dimensions, namely the effectiveness of physics-derived inversion and the physical self-consistency of model predictions—how different strategies capture the physical mechanisms underlying FMC. Table II summarizes the corresponding consistency metrics.

To examine the effectiveness of physics-derived inversion (first predicting EWT and LMA, then computing FMC), we used the model-predicted LMA and EWT traits to calculate FMC_phys via the definitional physical equation. We then evaluated the agreement between FMC_phys and the measured ground truth. The quantitative results illustrated in Table II and the scatter distributions in Fig. 5 show that FMC_phys, even when derived from LMA and EWT predicted by separate models (STL) (independent models) or the two-task model, remains highly consistent with observations ($R^2 > 0.91$) despite not being explicitly trained on an FMC task. Fig. 5 shows that the point cloud of FMC_phys is tightly clustered around the 1:1 line, confirming the robustness of the physical equation in trait inversion. However, this "high derivation accuracy" under STL or two-task learning masks a severe internal conflict: the SRD under the STL strategy is 4.71% and exhibits a pronounced long-tail distribution (Fig. 6). This indicates that although the independent models fit their respective targets well, the three outputs (LMA, EWT, and

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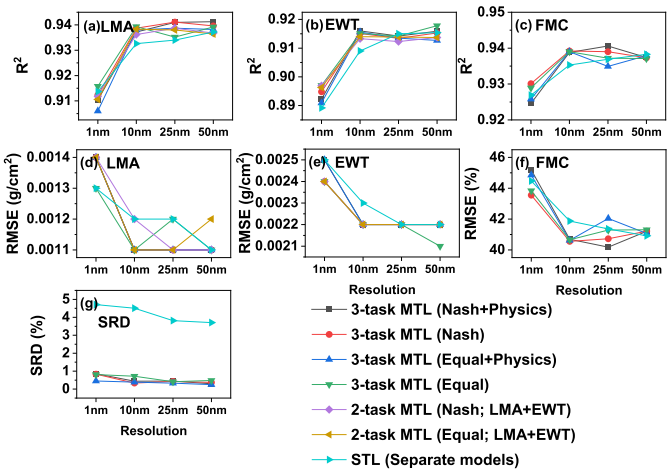



Fig. 7. Sensitivity analysis of model performance across varying spectral resolutions (1, 10, 25, and 50 nm). (a)–(c) Determination coefficients ($R^2$) for LMA, EWT, and FMC. (d)–(f) RMSE metrics for LMA, EWT, and FMC. (g) SRD represents physical self-consistency. Note: The 2-task MTL models are excluded from FMC and SRD comparisons (c), (f), and (g) as they do not directly predict FMC.

FMC) lack physical closure at the per-sample level. This results in a large systematic discrepancy between the directly predicted and physically derived values.

In terms of physical consistency, the SRD metric quantifies the difference between a model's direct outputs and values inferred through physical derivation. As demonstrated in Table II and Fig. 6, physics regularization significantly improved the model's intrinsic consistency. Introducing the physical constraint into the equal-weight MTL framework substantially decreased the mean SRD from 0.81% to 0.45%. This was accompanied by a more concentrated error distribution and a pronounced reduction in outliers. It is worth noting that improving physical consistency slightly reduces retrieval accuracy. After adding the regularization term, the $R^2$ of FMC_phys declined slightly (e.g., under Nash weighting, from 0.9286 to 0.9221). This phenomenon highlights the tension between data-driven fitting and physical-law constraints. Unconstrained, purely data-driven models (e.g., STL or unregularized MTL) tend to overfit noise in the training data to minimize the mean squared error. This disrupts the physical structure among variables. On the contrary, physics regularization guides the model toward solutions that are consistent with leaf physiological mechanisms. Although this limits the degrees of freedom available for fitting the observations, it reduces physically implausible predictions and systematic biases, ensuring greater physical coherence and robustness in multioutput settings.

### C. Robust Assessment Across Varying Spectral Resolutions

To evaluate the adaptability and robustness of the model across different spectral resolutions, the original 1-nm spectra were resampled into three broader-band settings (10, 25, and 50 nm). Then, all model variants were retrained and reevaluated. Fig. 7 shows how trait retrieval accuracy ($R^2$) and physical consistency (SRD) changed with spectral resolution.

Regarding the resolution dependence of retrieval accuracy [Fig. 7(a)–(f)], most models exhibit a consistent "increase-then-stabilize" pattern. Notably, performance at the original 1-nm resolution is slightly inferior to that at 10 nm and coarser settings. For example, $R^2$ is generally below 0.915 at 1 nm for LMA but increases to above 0.935 at 10 nm. This suggests that, while ultrafine hyperspectral resolution (1 nm) contains the richest information, it also introduces substantial high-frequency noise and spectral redundancy [57]. Moderate resampling (e.g., 10 nm) acts as a low-pass filter, improving the signal-to-noise ratio (SNR) and facilitating the extraction of robust spectral features by deep-learning models. Across the 10–50-nm range, model accuracy remains relatively stable. This indicates that the proposed MTL framework generalizes well to changes in spectral resolution and retains strong trait disentanglement capability under broader-band configurations. In terms of the physical consistency metric SRD [Fig. 7(g)], structural differences among model designs are particularly pronounced and stable. Regardless of spectral resolution (from 1 to 50 nm), the STL model (cyan line) maintains consistently high SRD values (>3.5%), indicating that physical inconsistency is an inherent limitation of this strategy that cannot be alleviated by reducing spectral dimensionality. On the contrary, the physics-regularized MTL models (black squares and blue triangles) maintain stable, very low SRD levels (<1%) across all resolutions. The equal weights model performs best. This provides strong evidence that the physical constraint mechanism is scale invariant. Regardless of the amount of spectral detail present in the inputs, the physics-aware network consistently preserves internal logical closure among the output variables (LMA, EWT, and FMC). This underscores its potential for multisource remote sensing data fusion. Finally, in a horizontal comparison of MTL strategies, the equal-weight MTL and Nash adaptive weighting strategies perform similarly across resolutions, with only minor differences. This again supports the conclusion in Section III-A that simple weighting strategies can be sufficiently robust for strongly related tasks.

## IV. Discussion

### A. Synergistic Retrieval of Leaf Traits via MTL

*1) Biophysical Trait Coupling and Mitigation of Error Propagation:* Our results indicated that although fundamental component traits such as LMA and EWT exhibited high retrieval stability across different model configurations (see Table I), the retrieval accuracy of the composite trait FMC was more sensitive to model structure. The three-task MTL architecture outperforms the two-task stepwise strategy, which "retrieves the component traits first and then computes FMC." This confirms that appropriately designed deep-learning frameworks can leverage biophysical coupling among traits to suppress error propagation. [19], [58]. Mechanistically, leaf spectral responses are not a simple linear superposition of individual traits; instead, they arise from nonlinear interactions among dry matter, water content, and internal leaf structure (as described, for example, by the PROSPECT model) [59], [60]. STL or two-task models tend to treat LMA and EWT as independent, orthogonal variables,

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thereby neglecting their coordinated variation along the leaf economics spectrum [61], [62], which in turn can lead to substantial biases in the derived FMC [63]. By contrast, the three-task setting explicitly includes FMC and uses a shared backbone to jointly encode, within a unified latent representation space, spectral features linked to SWIR water absorption (EWT-related) as well as to structural scattering and dry-matter information (LMA-related) [64]. This design forces the network to consider the covariance structure among traits when fitting each target individually. This "joint constraint" is a type of task-level regularization [37], [51] that improves the robustness of FMC estimation with small sample sizes and mitigates the error amplification of the stepwise strategy.

*2) Task Homogeneity and the Effectiveness of Weight Strategies:* When we compared task-weighting strategies, we found that the simple equal-weighting approach performed better overall than the computationally expensive Nash-MTL adaptive weighting scheme. Theoretically, Nash-MTL is primarily motivated by scenarios with pronounced multitask gradient conflicts, where tasks rely on mutually exclusive features and their gradient directions compete. This requires a bargaining-style update to obtain a fairer Pareto solution [29]. However, in leaf trait retrieval, LMA, EWT, and FMC are highly homologous in their physical definitions [3] and share overlapping spectrally sensitive regions, making task gradients more likely to be synergistic rather than strongly conflicting across most samples. In such cases, complex dynamic reweighting may not yield stable gains and can even become overly sensitive in later training stages to a small number of samples with large gradient-norm fluctuations (e.g., noisy or outlier points). This can introduce unnecessary perturbations to the shared representation. Based on this observation, our results supported the more pragmatic conclusion that, when retrieval tasks are physically well-coupled and homogeneous, maintaining a stable optimization objective (e.g., equal-weight linear scalarization) is more critical than pursuing sophisticated dynamic balancing. This conclusion aligns with the general understanding in multiobjective optimization that, when gradient conflicts are weak, simple scalarization can already yield effective solutions [65].

### *B. Tradeoff Between Accuracy and Consistency: Physics-Aware Modeling via Residual Correction*

This study revealed a high-accuracy illusion in purely data-driven models (STL). Although STL achieves excellent statistical accuracy for FMC ($R^2 > 0.91$), its SRD reaches as high as 4.71% (see Table II), indicating that the model may be fitting spectral noise or exploiting spurious correlations that happen to exist in the training set, rather than genuinely learning an interpretable physical relationship between FMC and leaf component traits [66], [67]. Such physical inconsistency can readily reduce model generalizability. After introducing physics regularization, we observed the expected tradeoff between the accuracy and consistency. While the $R^2$ of FMC slightly decreased, the SRD improved by an order of magnitude (down to 0.45%), which is consistent with the growing consensus on combining physical or mechanistic knowledge with data-driven learning [24], [68], [69]. From the perspective of residual learning, the proposed soft-physics constrained model can be interpreted as a "physical baseline + data-driven correction": (3) provides the physical baseline derived from EWT and LMA, and the residual correction term ($\delta$) denotes the residual correction term learned by the model [(4)].

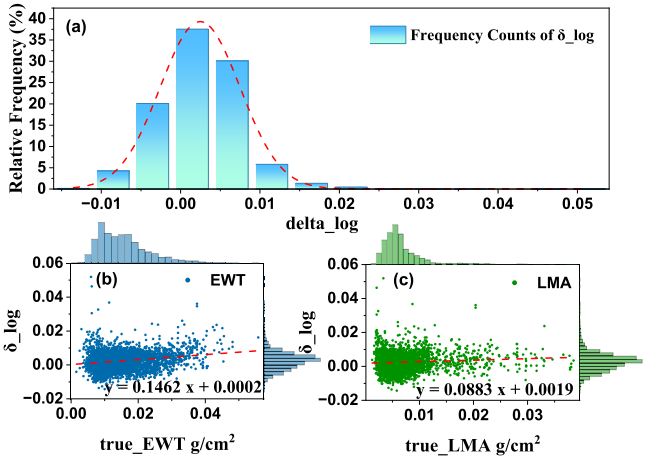



Fig. 8. Statistical analysis of the learnable residual correction term ($\delta$) in the physics-aware MTL model. (a) Frequency distribution histogram of delta with a Gaussian fit. (b) and (c) Scatter plots of delta against true EWT and true LMA, respectively.

Fig. 8 provides a detailed characterization of the correction term, $\delta$. Overall, $\delta$ exhibits a zero mean and a narrow distribution, which is consistent with the modeling assumption of a "baseline-dominated prediction with only minor residual adjustment" [Fig. 8(a)]. This suggests that the majority of the predicted FMC is influenced by the physical ratio of LMA and EWT. Only minor adjustments are permitted within a limited range. This helps to avoid the divergence in feature space that is often observed in purely data-driven models. Fig. 8(b) and (c) shows the joint distributions of $\delta$ with leaf state variables (EWT and LMA). Linear regression shows that $\delta$ forms diffuse, cloud-like scatter patterns with EWT and LMA, indicating weak correlations (with slopes of 0.146 and 0.088, respectively, showing a slight positive tendency). One plausible explanation is that $\delta$ primarily absorbs nonphysiological effects, such as interdataset discrepancies and inconsistent measurement errors, which are not strongly correlated with any single trait. However, the weak positive association between $\delta$ and EWT implies that the physical baseline may slightly underestimate FMC for high-water-content samples. From a spectral-mechanistic perspective, this could be attributed to nonlinear saturation in the SWIR water absorption features [70], [71]. Similarly, the weak positive relationship between $\delta$ and LMA may reflect the influence of the leaf structure parameter N [59]. Higher LMA is often associated with thicker leaves or denser mesophyll structures, which increase the path length that photons travel within the leaf by undergoing multiple scattering. This increases the difficulty of the inversion [72].

Overall, the slight decrease in FMC accuracy induced by physical-consistency regularization should not be interpreted

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as weakened model capacity. Rather, constraining the hypothesis space likely reduces the degrees of freedom available for fitting incidental correlations or high-frequency noise, in exchange for more consistent and more interpretable closure behavior.

### C. Generalization and Robustness: From Spectral Resolution to Ecological Domains

The sensitivity analysis across different spectral resolutions showed an unexpected finding: the original 1-nm spectra, which contain the richest spectral details, did not outperform the resampled broader-band data (10–50 nm) in retrieval accuracy. Instead, they performed slightly worse (Fig. 8). From the perspective of leaf optical mechanisms, this result is highly consistent with the absorption characteristics of LMA and EWT: leaf water (primarily around 970, 1200, 1450, and 1950 nm) [13], [59] and dry matter (mainly in the SWIR region) [14] are expressed largely as broad absorption features. While ultrafine 1-nm resolution can provide extremely subtle spectral fingerprints for deep-learning models, it also introduces substantial high-frequency measurement noise and strong spectral redundancy [57]. When the input dimensionality is excessively high and effective information gain is limited, models are prone to the curse of dimensionality and may be overfit to specific high-frequency noise patterns rather than true biophysical signals. Resampling to 10 nm effectively acts as an effective spectral smoothing filter, preserving the key broad absorption features and suppressing random noise to improve the SNR [18]. The quantitative results in supplementary tables S1–S3 further show that model performance remains stable across the 10–50-nm range. More importantly, we observed scale invariance in physical consistency (SRD). Regardless of how spectral resolution is degraded, as long as the spectra retain the basic absorption envelopes of water and dry matter, the physics-constrained model can produce solutions that satisfy the imposed constraint [26], [60]. This robustness provides a theoretical basis for multisource data fusion and suggests that achieving the finest possible spectral resolution is not always necessary.

Beyond the standard validation setting, this study further analyzed model behavior under extreme domain shifts. In particular, a stringent leave-one-dataset-out (LODO) stress test was conducted, in which the model was required to predict traits for PFTs or measurement protocols that had not been observed during training. In this strict zero-shot setting, we observed severe performance degradation in three subdatasets with markedly different spectral distributions from the training data. This caused the $R^2$ values of the single-task and equal-weight soft-physics multitask models to drop below zero. This finding confirms that data-driven models struggle to infer traits in domains with substantially different spectral characteristics in the absence of prior knowledge [11], [51]. Notably, the STL model exhibited reduced accuracy and substantially poorer physical consistency (SRD $\approx$ 14.72%, Supplementary Tables S4). On the contrary, the proposed equal-weight soft-physics multitask framework provided a more robust feasible-set constraint. Even when large domain discrepancies prevented accurate trait estimation, soft physical regularization effectively constrained the joint outputs of the three traits toward definitional closure. This maintained a very low physical violation rate (SRD $\approx$ 2.29%, Supplementary Tables S5). We explored the potential of few-shot calibration, which introduces 30% of the target-domain samples (Supplementary Tables S6 and S7), to mitigate the domain gap [73], [74]. In this setting, the equal-weight soft-physics multitask model achieved slightly lower $R^2$ values for EWT and LMA (approximately 0.70 and 0.65, respectively) than the STL model (approximately 0.71 and 0.71, respectively) while attaining a slightly higher $R^2$ value for FMC (approximately 0.57 versus approximately 0.55). More importantly, the physical violation rate of the equal-weight, soft-physics, multitask model (SRD $\approx$ 2.57%) remained substantially lower than that of the STL model (SRD $\approx$ 9.00%). These results suggest that the proposed framework cannot fully achieve "zero-shot universal inversion." However, under the more realistic assumption that a small amount of local calibration data is typically available, soft physical constraints can provide more reliable consistency guarantees and more controllable domain adaptation behavior.

Furthermore, it is worth noting that the composite traits discussed in this study are not limited to FMC, NOR to ratio forms. In ecological remote sensing and plant physiological research, there is a broad class of mixed indices with explicit analytical definitions, formed by multiple basic components (or intermediate variables) through deterministic calculation rules—for example, total chlorophyll (Chl a + b) [75], the vegetation carbon-to-nitrogen ratio (C/N) [76], or canopy water content (CWC) [77]. For these variables, the proposed strategy may provide a useful way to combine joint retrieval with internal-consistency constraints. However, the present study does not provide direct empirical validation for these cases. Their spectral sensitivities, coupling structures, and error-propagation characteristics may differ substantially from those of the LMA/EWT/FMC system considered here. Therefore, extending the framework to these traits can be regarded as a promising direction for future work.

## V. Conclusion

This study developed an MTL framework that integrates a TCN with soft physical constraints for the joint retrieval of leaf traits (LMA, EWT, and FMC), establishing a new paradigm of "physics-aware learning with data-driven correction." MTL achieves substantially higher physical consistency than single-task models. Furthermore, when tasks are strongly coupled and homogeneous, simple equal weighting can be more robust than complex, dynamic reweighting schemes (e.g., Nash-MTL). Consequently, the choice of optimization strategy in remote sensing inversion should be guided by the underlying physical synergy among target variables. Furthermore, the "physical baseline + residual correction" mechanism effectively mitigates the illusion of high accuracy in purely data-driven models by providing a favorable balance between physical consistency and predictive accuracy. This approach can easily be extended to other remote sensing variables that exhibit analytical coupling or conservation relationships. In addition, soft physical constraints enable the model to suppress high-frequency spectral noise more effectively, providing excellent

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stability across spectral resolutions. More importantly, the constraints establish a physics-based "safety floor" under extreme domain shifts, which helps reduce the risk of overconfident extrapolation by purely data-driven models in unseen environments.

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## REFERENCES

[1] Q. Yi, F. Wang, A. Bao, and G. Jiapaer, "Leaf and canopy water content estimation in cotton using hyperspectral indices and radiative transfer models," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 33, pp. 67–75, Dec. 2014.

[2] T. Cheng, B. Rivard, A. G. Sánchez-Azofeifa, J.-B. Féret, S. Jacquemoud, and S. L. Ustin, "Deriving leaf mass per area (LMA) from foliar reflectance across a variety of plant species using continuous wavelet analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 87, pp. 28–38, Jan. 2014, doi: 10.1016/j.isprsjprs.2013.10.009.

[3] J.-B. Féret et al., "Estimating leaf mass per area and equivalent water thickness based on leaf optical properties: Potential and limitations of physical modeling and machine learning," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 110959, doi: 10.1016/j.rse.2018.11.002.

[4] Q. Wang and P. Li, "Hyperspectral indices for estimating leaf biochemical properties in temperate deciduous forests: Comparison of simulated and measured reflectance data sets," *Ecolog. Indicators*, vol. 14, no. 1, pp. 56–65, Mar. 2012, doi: 10.1016/j.ecolind.2011.08.021.

[5] B. Fu et al., "Synergistic retrieval of mangrove vital functional traits using field hyperspectral and satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 131, Jul. 2024, Art. no. 103963, doi: 10.1016/j.jag.2024.103963.

[6] K. Helsen et al., "Evaluating different methods for retrieving intraspecific leaf trait variation from hyperspectral leaf reflectance," *Ecol. Indicators*, vol. 130, Nov. 2021, Art. no. 108111, doi: 10.1016/j.ecolind.2021.108111.

[7] R. Colombo et al., "Estimation of leaf and canopy water content in poplar plantations by means of hyperspectral indices and inverse modeling," *Remote Sens. Environ.*, vol. 112, no. 4, pp. 1820–1834, Apr. 2008, doi: 10.1016/j.rse.2007.09.005.

[8] M. M. Masinda, F. Li, L. Qi, L. Sun, and T. Hu, "Forest fire risk estimation in a typical temperate forest in Northeastern China using the Canadian forest fire weather index: Case study in autumn 2019 and 2020," *Natural Hazards*, vol. 111, no. 1, pp. 1085–1101, Mar. 2022, doi: 10.1007/s11069-021-05054-4.

[9] R. H. Nolan et al., "Drought-related leaf functional traits control spatial and temporal dynamics of live fuel moisture content," *Agricult. Forest Meteorol.*, vol. 319, May 2022, Art. no. 108941, doi: 10.1016/j.agrformet.2022.108941.

[10] B. Fu et al., "Cross-scenario transfer learning for estimating mangrove nitrogen and phosphorus content from field hyperspectral data to SDGSAT-1 and Sentinel-2 images," *Remote Sens. Environ.*, vol. 329, Nov. 2025, Art. no. 114923, doi: 10.1016/j.rse.2025.114923.

[11] G. Song, Y. Gan, L. Mao, Z. Ge, J. Lai, and Q. Wang, "Ensemble models enhanced estimates of leaf chlorophyll from fractional order derivatives transformed spectra," *Smart Agricult. Technol.*, vol. 12, Dec. 2025, Art. no. 101243, doi: 10.1016/j.atech.2025.101243.

[12] J. Jin and Q. Wang, "Informative bands used by efficient hyperspectral indices to predict leaf biochemical contents are determined by their relative absorptions," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 73, pp. 616–626, Dec. 2018, doi: 10.1016/j.jag.2018.08.002.

[13] H. Li, W. Yang, J. Lei, J. She, and X. Zhou, "Estimation of leaf water content from hyperspectral data of different plant species by using three new spectral absorption indices," *PLoS ONE*, vol. 16, no. 3, Mar. 2021, Art. no. e0249351.

[14] B. Yang, Y. He, and W. Chen, "A simple method for estimation of leaf dry matter content in fresh leaves using leaf scattering albedo," *Global Ecol. Conservation*, vol. 23, Sep. 2020, Art. no. e01201, doi: 10.1016/j.gecco.2020.e01201.

[15] C. He et al., "PROSPECT-GPR: Exploring spectral associations among vegetation traits in wavelength selection for leaf mass per area and water contents," *Sci. Remote Sens.*, vol. 8, Dec. 2023, Art. no. 100100, doi: 10.1016/j.srs.2023.100100.

[16] J. Sun et al., "Leaf pigment retrieval using the PROSAIL model: Influence of uncertainty in prior canopy-structure information," *Crop J.*, vol. 10, no. 5, pp. 1251–1263, Oct. 2022, doi: 10.1016/j.cj.2022.04.003.

[17] F. Vasseur et al., "A perspective on plant phenomics: Coupling deep learning and near-infrared spectroscopy," *Frontiers Plant Sci.*, vol. 13, May 2022, Art. no. 836488.

[18] W. Qi et al., "Intelligent retrieval of leaf traits using hyperspectral reflectance and deep learning," *Eur. J. Agronomy*, vol. 174, Mar. 2026, Art. no. 127960, doi: 10.1016/j.eja.2025.127960.

[19] E. Cherif et al., "From spectra to plant functional traits: Transferable multi-trait models from heterogeneous and sparse data," *Remote Sens. Environ.*, vol. 292, Jul. 2023, Art. no. 113580, doi: 10.1016/j.rse.2023.113580.

[20] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, Feb. 2019, doi: 10.1016/j.jcp.2018.10.045.

[21] S. Wang, X. Yu, and P. Perdikaris, "When and why PINNs fail to train: A neural tangent kernel perspective," *J. Comput. Phys.*, vol. 449, Jan. 2022, Art. no. 110768, doi: 10.1016/j.jcp.2021.110768.

[22] J. Verrelst et al., "Optical remote sensing and the retrieval of terrestrial vegetation bio-geophysical properties—A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 108, pp. 273–290, Oct. 2015, doi: 10.1016/j.isprsjprs.2015.05.005.

[23] Y. Zérah, S. Valero, and J. Inglada, "Physics-constrained deep learning for biophysical parameter retrieval from Sentinel-2 images: Inversion of the PROSAIL model," *Remote Sens. Environ.*, vol. 312, Oct. 2024, Art. no. 114309, doi: 10.1016/j.rse.2024.114309.

[24] P. Sun, P. M. van Bodegom, J. Timmermans, S. Liu, J. Wu, and M. D. Visser, "Hybrid modelling of leaf traits: Integrating neural networks with radiative transfer theory," *Remote Sens. Environ.*, vol. 329, Nov. 2025, Art. no. 114958, doi: 10.1016/j.rse.2025.114958.

[25] X. Quan, B. He, X. Li, and Z. Tang, "Estimation of grassland live fuel moisture content from ratio of canopy water content and foliage dry biomass," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1903–1907, Sep. 2015, doi: 10.1109/LGRS.2015.2437391.

[26] A. Daw, A. Karpatne, W. Watkins, J. Read, and V. Kumar, "Physics-guided neural networks (PGNN): An application in lake temperature modeling," 2017, *arXiv:1710.11431*.

[27] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7482–7491, doi: 10.1109/CVPR.2018.00781.

[28] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, "Gradient surgery for multi-task learning," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 5824–5836.

[29] A. Navon et al., "Multi-task learning as a bargaining game," 2022, *arXiv:2202.01017*.

[30] A. Khan, A. Sohail, U. Zahoora, and A. S. Qureshi, "A survey of the recent architectures of deep convolutional neural networks," *Artif. Intell. Rev.*, vol. 53, no. 8, pp. 5455–5516, Dec. 2020, doi: 10.1007/s10462-020-09825-6.

[31] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.

[32] H. Zhu, L. Yang, W. Ding, and Z. Han, "Pixel-level rapid detection of aflatoxin B1 based on 1D-modified temporal convolutional network and hyperspectral imaging," *Microchemical J.*, vol. 183, Dec. 2022, Art. no. 108020, doi: 10.1016/j.microc.2022.108020.

[33] C. Tu, W. Liu, W. Jiang, and L. Zhao, "Hyperspectral image classification based on residual dense and dilated convolution," *Infr. Phys. Technol.*, vol. 131, Jun. 2023, Art. no. 104706, doi: 10.1016/j.infrared.2023.104706.

[34] S. Kothari, M. Erding, and J. Cavender-Bares, "2018 Cedar creek pressed leaves," Cavender-Bares Lab, College Biol. Sci., Univ. Minnesota, St. Paul, MN, USA,Dataset, 2022, doi: 10.21232/b5uXd859.

[35] S. Kothari et al., "CABO 2018-2019 leaf-level spectra v2," Canadian Airborne Biodiversity Observatory, Université de Montréal, Montréal, QC, Canada, Dataset, 2022, doi: 10.21232/58kApoRy.

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[36] S. Kothari et al., "Dessain project reflectance spectra," Canadian Airborne Biodiversity Observatory, Université de Montréal, Montréal, QC, Canada,Dataset, 2022, doi: 10.21232/VYJzNBEy.
[37] T. Kattenborn, F. E. Fassnacht, and S. Schmidtlein, "Leaf reflectance plant functional gradient IFGG/KIT," Inst. Geography Geoecology (IFGG), Karlsruhe Inst. Technol. (KIT), Karlsruhe, Germany, Dataset, 2017, doi: 10.21232/dx1z-qz48.
[38] M. Grzybowski, N. K. Wijewardane, A. Atefi, Y. Ge, and J. C. Schnable, "Leaf spectra and physiological and chemical traits from maize grown under nitrogen stress," Schnable Lab, Dept. Agronomy Horticulture, Univ. Nebraska-Lincoln, Lincoln, NE, USA, Dataset, 2020, doi: 10.21232/y5TTxY3N.
[39] K. Helsen et al., "Leaf spectra of 4 plant species from Belgian dune grasslands + Rosa rugosa from the Northern Japan," Plant Conservation Population Biol. Lab, Dept. Biol., KU Leuven, Heverlee, Belgium, Dataset, 2021. Accessed: Sep. 20, 2025. [Online]. Available: https://ecosis.org/package/leaf-spectra-of-4-plant-species-from-belgian-dune-grasslands—rosa-rugosa-from-the-northern-japan
[40] S. P. Serbin and P. A. Townsend, "NASA FFT project leaf reflectance morphology and biochemistry for northern temperate forests," Brookhaven Nat. Lab., Upton, NY, USA, Dataset, 2014, doi: 10.21232/C2WC75.
[41] B. Hosgood, S. Jacquemound, G. Andreeoli, J. Verdebout, A. Pedrini, and G. Schmuck. (1993). *Leaf Optical Properties Experiment Database (LOPEX93)*. Dataset. Accessed: Sep. 20, 2025. [Online]. Available: http://opticleaf.ipgp.fr/index.php?page=database
[42] S. Jacquemound, L. Bidel, C. Francois, and G. Pavan. (2003). *ANGERS Leaf Optical Properties Database*. Ecological Spectral Information System (EcoSIS), Dataset. Accessed: Sep. 20, 2025. [Online]. Available: https://ecosis.org/package/angers-leaf-optical-properties-database–2003-
[43] B. Zimmermann and A. Kohler, "Optimizing Savitzky–Golay parameters for improving spectral resolution and quantification in infrared spectroscopy," *Appl. Spectrosc.*, vol. 67, no. 8, pp. 892–902, Aug. 2013, doi: 10.1366/12-06723.
[44] G. Morales, J. W. Sheppard, R. D. Logan, and J. A. Shaw, "Hyperspectral dimensionality reduction based on inter-band redundancy analysis and greedy spectral selection," *Remote Sens.*, vol. 13, no. 18, p. 3649, Sep. 2021, doi: 10.3390/rs13183649.
[45] R. J. Barnes, M. S. Dhanoa, and S. J. Lister, "Standard normal variate transformation and de-trending of near-infrared diffuse reflectance spectra," *Appl. Spectrosc.*, vol. 43, no. 5, pp. 772–777, Jul. 1989. [Online]. Available: https://opg.optica.org/as/abstract.cfm?URI=as-43-5-772
[46] S. Ruder, "An overview of multi-task learning in deep neural networks," 2017, *arXiv:1706.05098*.
[47] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9, doi: 10.1109/CVPR.2015.7298594.
[48] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
[49] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
[51] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
[52] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*.
[53] P. Borah and D. Gupta, "Functional iterative approaches for solving support vector classification problems based on generalized Huber loss," *Neural Comput. Appl.*, vol. 32, no. 13, pp. 9245–9265, Jul. 2020, doi: 10.1007/s00521-019-04436-x.
[54] D. Q. F. de Menezes, D. M. Prata, A. R. Secchi, and J. C. Pinto, "A review on robust M-estimators for regression analysis," *Comput. Chem. Eng.*, vol. 147, Apr. 2021, Art. no. 107254, doi: 10.1016/j.compchemeng.2021.107254.
[55] T.-T. Wong and P.-Y. Yeh, "Reliable accuracy estimates from k-fold cross validation," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 8, pp. 1586–1594, Aug. 2020, doi: 10.1109/TKDE.2019.2912815.
[56] R. Nuzzo, "Percent differences: Another look," *PM&R*, vol. 10, no. 6, pp. 661–664, Jun. 2018, doi: 10.1016/j.pmrj.2018.05.003.
[57] J. Jia et al., "The effect of artificial intelligence evolving on hyperspectral imagery with different signal-to-noise ratio, spectral and spatial resolutions," *Remote Sens. Environ.*, vol. 311, Sep. 2024, Art. no. 114291, doi: 10.1016/j.rse.2024.114291.
[58] P. Farajpoor, A. Pourreza, M. Narimani, A. El-kereamy, and M. W. Fidelibus, "Multi-trait spectral modeling for estimating grapevine leaf traits and nutrients," *Plant Phenomics*, vol. 7, no. 4, Dec. 2025, Art. no. 100142, doi: 10.1016/j.plaphe.2025.100142.
[59] S. Jacquemoud and F. Baret, "PROSPECT: A model of leaf optical properties spectra," *Remote Sens. Environ.*, vol. 34, no. 2, pp. 75–91, Nov. 1990, doi: 10.1016/0034-4257(90)90100-z.
[60] J.-B. Féret, A. A. Gitelson, S. D. Noble, and S. Jacquemoud, "PROSPECT-D: Towards modeling leaf optical properties through a complete lifecycle," *Remote Sens. Environ.*, vol. 193, pp. 204–215, May 2017, doi: 10.1016/j.rse.2017.03.004.
[61] Z. Wang et al., "Leaf water content contributes to global leaf trait relationships," *Nature Commun.*, vol. 13, no. 1, p. 5525, Sep. 2022, doi: 10.1038/s41467-022-32784-1.
[62] I. J. Wright et al., "The worldwide leaf economics spectrum," *Nature*, vol. 428, no. 6985, pp. 821–827, Apr. 2004, doi: 10.1038/nature02403.
[63] L. Zhu, G. I. Webb, M. Yebra, G. Scortechini, L. Miller, and F. Petitjean, "Live fuel moisture content estimation from MODIS: A deep learning approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 81–91, Sep. 2021, doi: 10.1016/j.isprsjprs.2021.07.010.
[64] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, Jul. 1997, doi: 10.1023/a:1007379606734.
[65] O. Sener and V. Koltun, "Multi-task learning as multi-objective optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–11.
[66] W. Ye, G. Zheng, X. Cao, Y. Ma, X. Hu, and A. Zhang, "The clever Hans mirage: A comprehensive survey on spurious correlations in machine learning," 2024, *arXiv:2402.12715*.
[67] O. Salaudeen, H. Zhang, K. Alhamoud, S. Beery, and M. Ghassemi, "Aggregation hides out-of-distribution generalization failures from spurious correlations," 2025, *arXiv:2510.24884*.
[68] C. Meng, S. Griesemer, D. Cao, S. Seo, and Y. Liu, "When physics meets machine learning: A survey of physics-informed machine learning," *Mach. Learn. Comput. Sci. Eng.*, vol. 1, no. 1, p. 20, Jun. 2025, doi: 10.1007/s44379-025-00016-0.
[69] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Comput. Surv.*, vol. 55, no. 4, pp. 1–37, Apr. 2023.
[70] P. Ceccato, N. Gobron, S. Flasse, B. Pinty, and S. Tarantola, "Designing a spectral index to estimate vegetation water content from remote sensing data: Part 1," *Remote Sens. Environ.*, vol. 82, nos. 2–3, pp. 188–197, Oct. 2002, doi: 10.1016/s0034-4257(02)00037-8.
[71] H.-D. Seelig, A. Hoehn, L. S. Stodieck, D. M. Klaus, W. W. Adams, and W. J. Emery, "Relations of remote sensing leaf water indices to leaf water thickness in cowpea, bean, and sugarbeet plants," *Remote Sens. Environ.*, vol. 112, no. 2, pp. 445–455, Feb. 2008, doi: 10.1016/j.rse.2007.05.002.
[72] E. G. de la Riva, M. Olmo, H. Poorter, J. L. Ubera, and R. Villar, "Leaf mass per area (LMA) and its relationship with leaf structure and anatomy in 34 Mediterranean woody species along a water availability gradient," *PLoS ONE*, vol. 11, no. 2, Feb. 2016, Art. no. e0148788, doi: 10.1371/journal.pone.0148788.
[73] F. Ji et al., "Leveraging transfer learning and leaf spectroscopy for leaf trait prediction with broad spatial, species, and temporal applicability," *Remote Sens. Environ.*, vol. 326, Aug. 2025, Art. no. 114818, doi: 10.1016/j.rse.2025.114818.
[74] T. U. Rehman and J. Jin, "Deep adversarial domain adaptation for hyperspectral calibration model transfer among plant phenotyping systems," *Biosystems Eng.*, vol. 224, pp. 246–258, Dec. 2022, doi: 10.1016/j.biosystemseng.2022.10.016.
[75] B. Datt, "Remote sensing of chlorophyll a, chlorophyll b, chlorophyll a+b, and total carotenoid content in eucalyptus leaves," *Remote Sens. Environ.*, vol. 66, no. 2, pp. 111–121, Nov. 1998, doi: 10.1016/s0034-4257(98)00046-7.
[76] A. Ostrowska and G. Porębska, "Assessment of the C/N ratio as an indicator of the decomposability of organic matter in forest soils," *Ecolog. Indicators*, vol. 49, pp. 104–109, Feb. 2015, doi: 10.1016/j.ecolind.2014.09.044.
[77] J. Liu et al., "Assessment of mango canopy water content through the fusion of multispectral unmanned aerial vehicle (UAV) and Sentinel-2 remote sensing data," *Forests*, vol. 16, no. 1, p. 167, Jan. 2025, doi: 10.3390/f16010167.

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